

RAGHAV SAREEN

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OBJECTIVE

Computer Engineering student seeking Firmware Engineering opportunities, with hands-on experience in embedded C/C++, STM32 firmware, motor control, and edge-AI robotics.

EDUCATION

Bachelor of Science in Computer Engineering, San Jose State University

Expected May 2028

GPA: 3.776/4.000 | 2026 President's Scholar

Selected Coursework: Object-Oriented Programming, Electronics, Algorithms and Data Structures, Assembly Language Programming

SKILLS

Languages C, C++, Python

Embedded Systems STM32F103, libHAL, GPIO, CAN, PWM, I²C, PID control, BLDC motors

Platforms & Tools NVIDIA Jetson Orin Nano, Linux, Git/GitHub, Docker, Tailscale, Jira

AI & Computer Vision OpenClaw, Claude Code, ChatGPT, Ultralytics YOLO, TensorRT, OpenCV

TECHNICAL EXPERIENCE

Firmware Team Member, ALICANTO Rover

Sep 2024 – May 2026

SJSU Robotics

San Jose, CA

- Programmed and tested C++ firmware for the ALICANTO rover arm subsystem for the 2027 University Rover Challenge; committed changes that were reviewed and merged into the team codebase.
- Validated three STM32F103 MicroMod and motor assemblies controlling the wrist, shoulder, and elbow, verifying motor response and component behavior with libHAL.
- Tuned PID control functions to reduce arm jitter and improve stability; implemented arm homing for BLDC motors using a limit switch, GPIO, CAN, and PWM.

AI Agent Testing Volunteer

Jun 2026

Social Experiments

Remote

- Designed 60 functional test cases and executed 50 functional and end-to-end scenarios for an AI Slack agent spanning Jira integration, contract drafting, ingestion, redlining, approvals, and permissions.
- Documented preconditions, steps, expected versus actual results, priorities, and defects; identified failures or errors in 28 scenarios, including duplicate responses, timeouts, and incorrect task routing.
- Shared findings with the product team, contributing to improvements across most identified workflows; applied Jira, Kanban, Scrum, and PRD concepts including epics, features, and user stories.

PROJECTS

AI-Assisted Home Security Arm Turret | *Jetson Orin Nano, Docker, YOLO, TensorRT, Tailscale* Jul 2026 – Present

Conceived, assembled, deployed, and validated a 24/7 edge-AI security system integrating a USB camera and two xArm servos for pan/tilt person tracking. Used OpenClaw and Claude Code for AI-assisted implementation; configured and tested automatic scanning, face verification, target reacquisition, person/dog modes, Telegram alerts, and an authenticated control dashboard.

The Efficacy of ChatGPT in Mathematics | *Polygence Research Program*

May 2023 – Sep 2023

Evaluated ChatGPT on 107 algebra problems, measuring 81% initial accuracy and comparing follow-up prompting strategies. Analyzed failure patterns and tested LaTeX and role prompting for clearer responses; authored a research paper and won Top Asynchronous Pitch among 300+ symposium presentations.

LEADERSHIP & ACTIVITIES

Digital Content Creator

Jun 2020 – Present

Built and managed a pet-focused brand across Instagram, YouTube, and TikTok, reaching 30,000+ Instagram followers, 1,700+ YouTube subscribers, and 1.6M+ combined views.